Chapters

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Intro.

In this document, I will explore the concept of counting to twelve as a single unit rather than counting to ten. This paper is written for a general audience rather than industry experts. For professionals in fields related to this topic, I welcome your expertise. I invite you to view this paper not as an absolute truth, but as an open invitation to look beyond existing dogmas or narrow perspectives.

This document is an offshoot from a bigger part and is wholly focussed on duodecimal counting. (see, https://github.com/MadHatter-Auditor/vortex-anapole-confinement/blob/main/Base-12%20Microcosm.pdf.) Because of this some chapters will be recognizable from that bigger part as also some parts will be further elaborated on.

Base-12 System Introduction.

Normally the duodecimal system is written down as this: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, 10. In this notation 9=nine, A=ten, B=eleven, 10=dozen, 11=dozen-one, and 12=dozen-two. This so that 13=dozen-three, etcetera.

So why even dozen and why this linguistic lingo?

So, the problem starts with how many numbers we have. We don't have more than 10 numbers counting from 0 to 9. So, you have to add two numbers that we don't have. Then you also have the problem that the 10 does not have its own number but is rather a combination of the first value added with a 0 to define that we have a whole round. 10.20.30 is the same as 1.2.3 but now times 10. So, if we let 10 be 10 then what is 11? Then the problems only add up from there so they invented a whole linguistic system to make sense of how it works.

But ironically the problem lies with **thinking** in base-10 at a linguistic level. I'll be honest and spare the reader a lot of thinking time.

This is an outdated system and we will drop it. Instead, we will use 0,1,2,3,4,5,6,7,8,9,α,ω,10. This seems like a small change but linguistically this will work much easier. All the numbers stay the same in language but we add α and ω as letters called Alph and Ohm. This way we can count nine, alph, ohm, ten eleven and twelve as 9,α,ω,10,11,12. This way larger numbers also work, such as α00 and ω00 as Alph-hundred and Ohm-hundred. Or even αα Alphty Alph. And ααα would then be Alph-hundred Alphty Alph. Notice that the exact rules of base-10 counting can work on base-12 where numbers in ten have linguistically ''ty'' behind it. the first and last letters of the Greek alphabet felt fitting since I only needed two new symbols and they are the first and last from that alphabet.

I specifically chose these Greek symbols because we already use Roman letters in mathematics, and I wanted to avoid making regular numbers look like algebraic formulas.

Base-10		Base-12	
0	0	0	0
1	1	1	1
	2	2	
	3	3	
	4	4	
	5	5	
	6	6	
	7	7	
	8	8	
	9	9	
	10	α	
2	11	ω	2
	12	10	
	13	11	
	14	12	
	15	13	
	16	14	
	17	15	
	18	16	
	19	17	
	20	18	
3	21	19	3
	22	1α	
	23	1ω	
	24	20	
	25	21	
	26	22	
	27	23	
	28	24	
	29	25	
	30	26	
4	31	27	4
	32	28	
	33	29	
	34	2α	
	35	2ω	
	36	30	
	37	31	
	38	32	
	39	33	
	40	34	
5	41	35	5
	42	36	
	43	37	
	44	38	
	45	39	
	46	3α	
	47	3ω	
	48	40	
	49	41	
	50	42	
6	51	43	6
	52	44	
	53	45	
	54	46	
	55	47	
	56	48	
	57	49	
	58	4α	
	59	4ω	
	60	50	

On the right I have made a visualisation.
I have put on every number now also a colour. On the left we have base-10 and on the right base-12. This visual representation should provide a clear understanding of the shift per group. On the left the group is 10 but on the right the group is as big as the 12 on the left. This means that a per ten in duodecimal means 12 while in denary (base-10) it is simply ten. This is something that is present in every base system besides denary.

I made it to 60 to illustrate that that in base-10 you need to do 6 times and in base-12 only 5 times.

Greek Duodecimal System (GDS).

In base-12, multiple notations exist. While I used the computer notation as the first example, there are also the Pitman and Dwiggins notations, which utilize symbols specifically designed for duodecimal counting. Furthermore, there is the telephone notation, briefly used in the 1970s and 1980s, which features the * and # symbols.

However, each system has major flaws. The computer notation clashes with other fields of mathematics because A and B are already heavily utilized in algebraic formulas. Pitman and Dwiggins use symbols that are not universally recognized; designing notation that is not standard in digital systems creates severe software compatibility issues, requiring decades of lobbying to get custom characters adopted. Lastly, the telephone notation proved practically unworkable because the symbols resemble punctuation rather than values and conflict with existing programming and mathematical functions.

Because the letters used are Greek, the notation is called the Greek Duodecimal system, in short GDS.

This notation of lowercase α and ω solves multiple problems. Even though ω is used in electronics its notation is Ω and an uppercase α is simply the same as the Latin ‘’A’’. This means that it does not collide with other mathematics. The clash I do see with electronics is linguistically but Ohm Ohm’s could become rather a little joke than a big problem knowing mechanics. The same could be said about what and watt. the choice to leave all other numbers exactly the same in language is a massive advantage that the other notations don’t utilize.

The biggest reason for leaving as much intact as possible is due to how people adapt. Duodecimal counting is something people are not used to, and using familiar letters while leaving the inner linguistic structure intact will prove to be far more understandable than introducing an entirely new linguistic framework. In short, convincing people that 10 has a value of twelve is challenging enough; inventing completely new vocabulary is an additional obstacle that can (and should) be avoided.

There are however some other problems that remain present. particularly regarding angles and angular velocity. In physics and geometry, this notation could lead to visual paradoxes such as α = α0◦ or ω=ω rad/s. If it is clearly stated beforehand that the notation is duodecimal, it technically poses no structural mathematical problem. However, it could still easily cause confusion

for a fast reader who is accustomed to treating these symbols strictly as variables rather than literal numerical values."

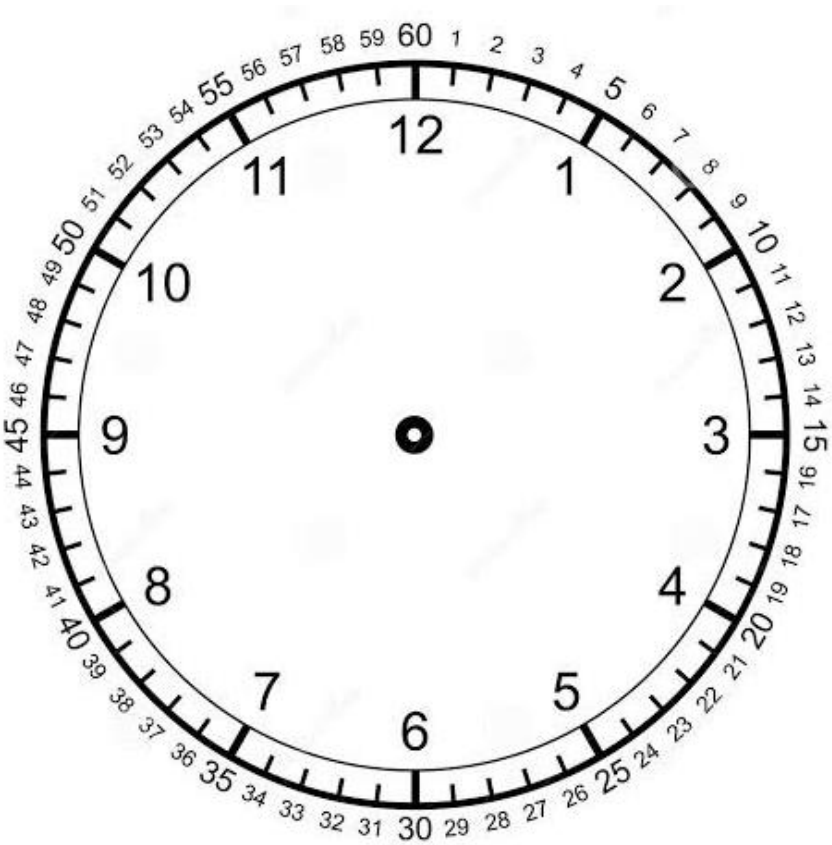
Other notable challenges include Cantor's ω and the linguistic shift in 10 and 12. The first one is because ω is the first infinite number after all normal finite numbers. So, when counting with infinites this could pose a problem. The other part is just a mental part of when we see 10 and we think of the value in base-10 while the amount would be 12. Another thing but workable is the notion of working with powers like 10^{-9} with α this can be written down as 10^-α. To put it in a row. 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}, 10^{-9}, 10^-α 10^-ω and 10^{-10}. On this part I got lucky with the Caret-notation that is used sometimes instead of superscript. Both are known so it is a stylistic choice to either use only caret or a combination like the example given.

Duodecimal time and dimensions.

But why choose a base-12 system? Base-10 is the golden standard, and everyone seems to agree it is an excellent system. Well, yes, it is a good system, but it is also a limited one. Ten is only easily divisible by 2 and 5, whereas twelve can be divided by 2, 3, 4, and 6. In daily activities like cooking or checking the time, we use quarters and thirds far more often than fifths.

This brings me to the next point and topic of this chapter, the clock. Why do we use base-12 for measuring hours but revert to base-10 for minutes? If we applied a base-12 logic to minutes, the relative value of time would remain the same, but the numerical grid would perfectly align with the hours.

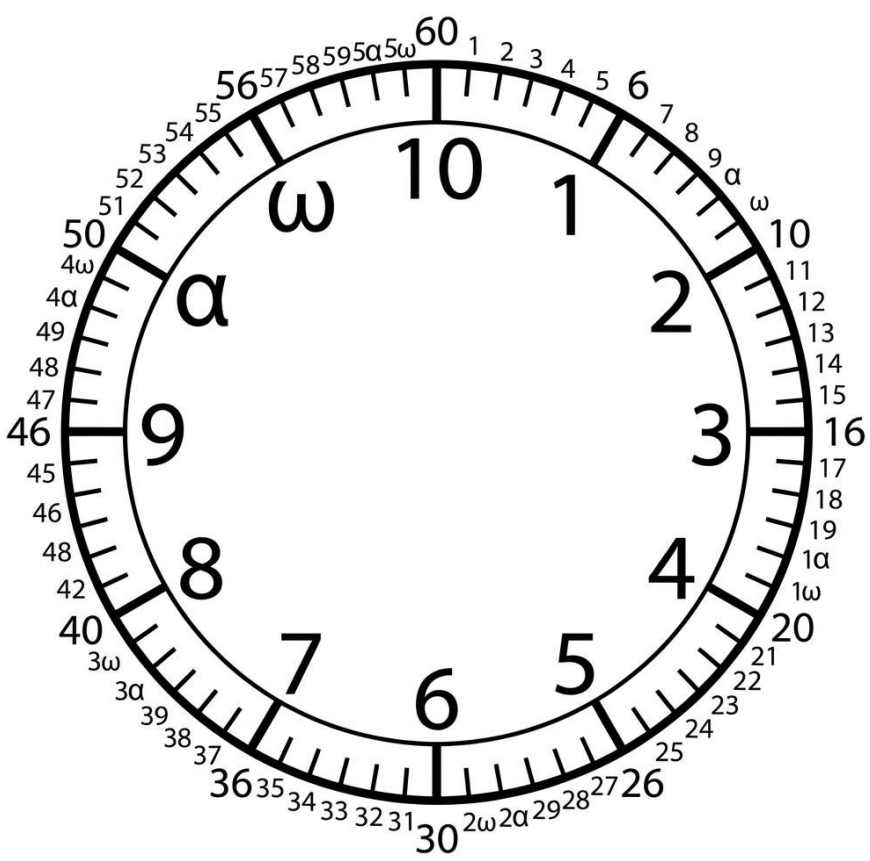
Let's have a closer look to the normal clock. We can see the division between the hours and the minutes. The hours here are in base-12
24 hours a day and 12 half a day. The minutes are in base-10. 15 minutes quarter, 30 half, 45 threequarters and 60 a full hour.



So, we count the quarters of the day in steps of 3 and the quarters of an hour by steps of 15. That means that during the day we constantly switch between the table of 5 and the table of 3.

You could already spot the mismatch when looking at the part between the 12 a clock (representing 0,) and the 1 a clock. Put a minute hand halfway though. Half of 5 minutes is 2.5 minutes but half of the clock stands on 6 hours or 30 min. If they would be the same there would be 3 min instead of 2.5 min.

Now let's have a look at a clock according to GDS. Instead of 60 minutes we have 72 minutes even though we would still call that 60 minutes.



Here, we can observe that the hour markings seamlessly match the progression of the minutes and seconds. The minute marks are no longer numerically decoupled from the hour marks. For instance, when we look at the 2 o'clock mark, it aligns perfectly with the **10**-minute position. In traditional denary terms, this signifies 12 units, which is exactly the cumulative value of two completed sectors. Formerly, this structural point on a decimal face would read as 10 minutes, completely failing to correlate with the macro value of the two hours.

This visualization also demonstrates the mathematical consistency of the duodecimal positional shift. If you multiply any hour position by 6, it yields the exact unit mark on the outer ring:

- **In Duodecimal (GDS):** α x 6 = 50 and ω x 6 = 56.
- **In Denary (Base-10):** 10 x 6 = 60 and 11 x 6 = 66.

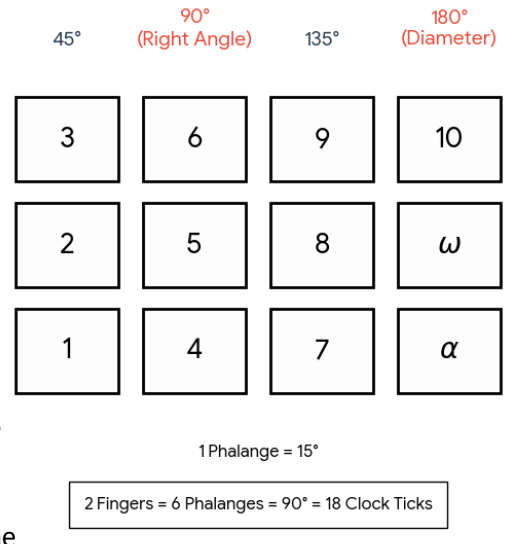
The transitional difference between the values spans exactly through 5α (fifty-alph) and 5ω (fifty-ohm), totalling to the next position layer. Consequently, it is no surprise that the final sum, incorporating these blocks, lands precisely on a total value of **60**, which correlates perfectly to **72** units in denary notation. Compared to denary the shift over 12 is always 12.

This would mean some revision on some time conversion methods but since we already measure with hours such a revision would be more of a streamlining rather than a total rework of known science. For example, in a standard hour conversion. Where we normally calculate a quarter-hour as 15 minutes × 60 seconds = 900 seconds, the GDS grid scales differently. Under this system, a quarter-cycle is calculated as 3 hours × 100 GDS minutes, cleanly yielding 300 GDS minutes (the exact equivalent of those 900 traditional seconds). In this way, we elegantly shift our daily time-tracking from a messy "minutes × seconds" calculation to a much more intuitive "hours × minutes" grid.

Sidenote, the hour now also gives away the minutes, 2 hours is 100 GDS minutes (120 units in denary)

Duodecimal in practice.

Where we look at a duodecimal clock, I think we should leave time at that. To go and recount history into duodecimal would be absurd and useless. To make the clock correlating with 360° and the hours, minutes and seconds congruent doesn't mean the whole timeline should be in duodecimal, there is a limit and a sweet spot for practicality.

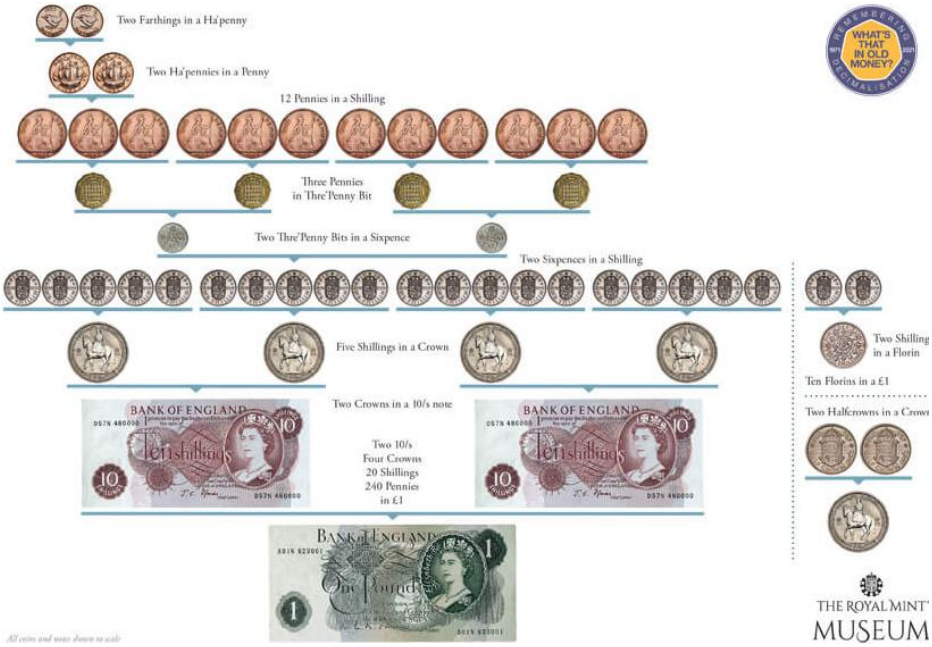


Luckily it is the adding of data and not a subtraction of data so by that means it can be an upgrade where needed and let alone where not needed. The hand is used as a counting tool to this very day; however, instead of counting the fingers themselves, for GDS you can look at the phalanges.

With the thumb you can count now not only the amount but also the angles. one finger is 45° and one phalanges is 15°.
6 phalanges =90° = 16 minutes so 1 phalanges = 3 minutes.
Not only can you count to ten but rather tell time, tell angle and calculate with more numbers than normally.

This is no new science though, the counting with phalanges came from the ancient Egyptians. And even though duodecimal counting is in modern day is almost unheard of this was certainly not the case for the most part of human history. The ancient Babylonians came up with the base-60 system where our modern hour notation comes from. And even not so long ago it was used in common everyday life, money.

Let us examine the history of the shilling. This diagram, sourced from the Royal Mint Museum, explains it clearly.



To demonstrate why this is such an elegant system, let us look at a practical scenario. If I go out with two friends and a beer costs 2 shillings, we can perfectly divide that total by three, meaning each person pays exactly 8 pence.

In modern times, if a beer costs €2.40 and we divide it among three people, each person owes €0.80. However, if the total bill is exactly €1.00 and we split it three ways, we get an endless decimal of €0.3333... per person. The pre-decimal British currency system elegantly avoided this issue.

Below is a short comparison demonstrating how easily different numbers of people can clean-split a single unit of currency:

Divided by:	Euro (€1.00/100 cent)	Shilling (12pence)	Clean numbers
2 persons	50 cents	6 pence	Both
3 persons	33.33 cent	4 pence	Shilling
4 persons	25 cents	3 pence	Both
5 persons	20 cents	2.4 pence	Euro
6 persons	16.66 cent	2 pence	Shilling
12 persons	8.33 cent	1 penny	Shilling

This raises an important question: in daily life, how often do we need to divide something by 5, compared to dividing it by 3, 4, 6, or 12? While computers handle repeating decimals effortlessly, humans frequently accumulate significant rounding errors over time when using a base-10 system. Furthermore, modern prices like €99.99 are unnecessary and psychologically manipulative, a practice best described as "profit through obscurity."

Adopting a base-12 system does not eliminate decimals entirely; unconventional fractions will always exist. However, because a base-12 system offers more factors, the occurrence of infinitely repeating decimals is drastically reduced.

Take the fraction 1/27 as an example. In base-10, this results in the endless sequence 0.037037037.... In a base-12 system, the exact same fractional value resolves cleanly to a terminating expression: 0.054 (written here using base-10 fractional values to avoid confusion, though in pure base-12 notation, the fraction itself would be written as 1/23).

With practicality in mind, I also come back to the point about regarding angles, angular velocity and Cantor's ω. In the case of programming there could be the option to use variant characters like Var omega or some other slight variant. This is not as much of a problem as long as a person is considered but when it comes to parsing data with computers this could be a big problem if not implemented correctly. $y{=}\alpha d{\cdot}\omega d$ or $y{=}\#\alpha{\cdot}\#\omega$ are also probable solutions or even more simply, α *(by U+1D6FC)* = α (by U+03B1) 0∘.

Ironically, we still use duodecimal counting to this day. The parts that did hold up are the most practical things. Let's say you go on groceries. How much beers are in one crate? What is a 6-pack? And ever looked at the eggs? The word dozen duodecimal society uses literally comes from 12 eggs. The reason is not only by how many you can divide but also weight division like with a 6-pack. People avoid 6-packs with one missing more often than with 2 missing or they even put one back.

Basic GDS arithmetic.

With the focus on psychology, I decided to make an overview of basic arithmetic to show from the start up.

With the added colours we can see even more clearly the patterns. The base-10 is in that case literally a narrower view of arithmetic. In basic instinct one would say that that is less work and easier to understand. But that is not quite as true as the ‘‘amount’’ (numbers) of tables to learn is the same.

With denary we need the tables of
3,4,7 and 8
With duodecimal we need the tables of
5,7,9 and α

These are the drill tables and bare minimum of both systems to understand and calculate the others.

The other tables as one can see are quite easily remembered when you know the pattern.

denary (Base-10)											
1	2	3	4	5	6	7	8	9	10	11	12
2	4	6	8	10	12	14	16	18	20	22	24
3	6	9	12	15	18	21	24	27	30	33	36
4	8	12	16	20	24	28	32	36	40	44	48
5	10	15	20	25	30	35	40	45	50	55	60
6	12	18	24	30	36	42	48	54	60	66	72
7	14	21	28	35	42	49	56	63	70	77	84
8	16	24	32	40	48	56	64	72	80	88	96
9	18	27	36	45	54	63	72	81	90	99	108
10	20	30	40	50	60	70	80	90	100	110	120
11	22	33	44	55	66	77	88	99	110	121	132
12	24	36	48	60	72	84	96	108	120	132	144

GDS (base-12)											
1	2	3	4	5	6	7	8	9	α	ω	10
2	4	6	8	α	10	12	14	16	18	1α	20
3	6	9	10	13	16	19	20	23	26	29	30
4	8	10	14	18	20	24	28	30	34	38	40
5	α	13	18	21	26	2ω	34	39	42	47	50
6	10	16	20	26	30	36	40	46	50	56	60
7	12	19	24	2ω	36	41	48	53	5α	65	70
8	14	20	28	34	40	48	54	60	68	74	80
9	16	23	30	39	46	53	60	69	76	83	90
α	18	26	34	42	50	5α	68	76	84	92	α0
ω	1α	29	38	47	56	65	74	83	92	α1	ω0
10	20	30	40	50	60	70	80	90	α0	ω0	100

The table of 9 is famous in denary for its easy implementation, where the digits adapt through simple addition and subtraction. In duodecimal, it is just as straightforward, as the final digit alternates predictably between 9, 6, 3, and 0. Furthermore, the table of 11 (ω) inherits the exact same countdown trick that the table of 9 possesses in denary. The table of 10 (α) operates similarly, decreasing by steps of two on the right digit while increasing by one on the left. Consequently, these additional tables are remarkably easy to master once their underlying patterns are understood.

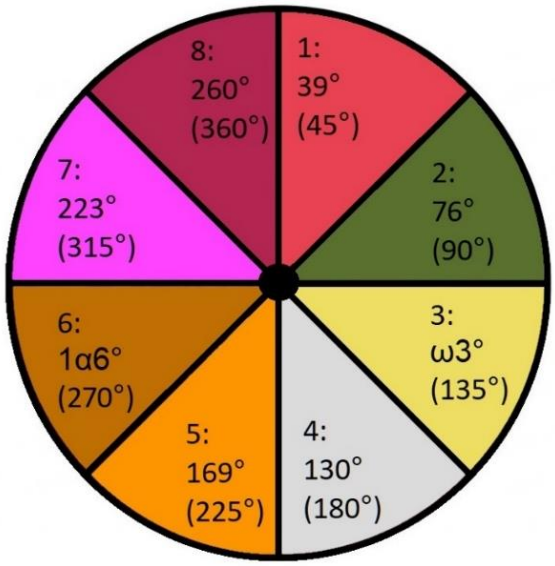
GDS Degrees.

Normally, a full circle is divided into 360 degrees. I chose to leave the physical quantity of this 360-degree circle intact, as changing the physical baseline would yield little practical benefit. However, by upgrading the numerical interface to GDS notation, the mathematical grid becomes far more harmonious. Here is a short overview of the traditional quadrant angles translated into GDS notation: 45° (denary) = 39° (GDS), 90° = 76°, 135° = ω3°, 180° = 130°, 225° = 169°, 270° = 1α6°, 315° = 223° and 360° = 260°. As one might notice, the tables of 5 and 9 are utilized here, which makes degrees one of the hardest components to instinctively grasp. Ironically, while these tables represent the easiest, most intuitive patterns in denary, they transition into the most challenging drill tables within the duodecimal framework.

This is also a great example of the sweet spot of practicality. Where duodecimal is not needed it should be better to firstly avoid. On the hand-map shown earlier I mapped the degrees in normal base-10 with a straight angle of 90° rather than the GDS 76° number. This is the same way if you are working with royal cubits as measurement. You just adapt the data to the most practical application and in that case would be the metric system. The reasoning is that when making blueprints and measurements it would be strange to see a straight angle that says GDS 76°, an engineer would also not recognize 1α6° without it being clearly stated that the documentation is on GDS. If GDS is applied to angles than it would be easier when basic arithmetic is also in duodecimal. Luckily this is only a matter of documentation and switching to 260° would not break any natural laws since the amount stay's exactly the same. This is also why this is a good point to capitulate to practicality on this matter. This can easily be ISO defined if needed.

Even though I did decide to make charts to visualize what's happening.

Here we can see GDS divided over 8 parts. We can also see a clear characteristic of denary with a clear preference in the table of 2 rather than the table of 3 that duodecimal uses. Below in short, a summary.



39°, 76°, ω3°, 130°, 169°, 1α6°, 223° and 260°.

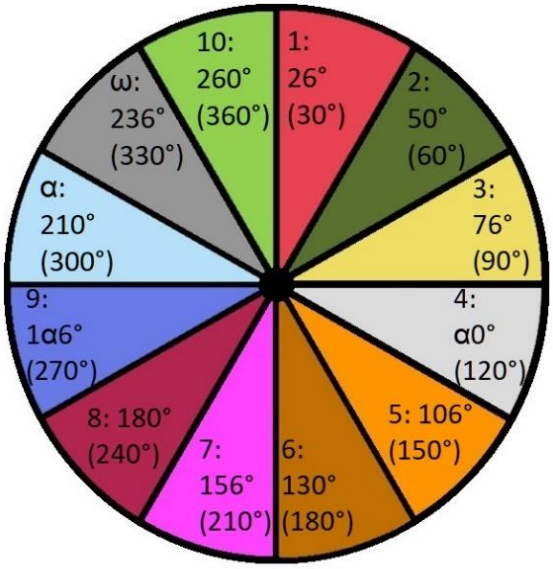
But now instead of looking at the circle with the normal 8 parts as above, what happens when we look at duodecimal? we get this.

26°, 50°, 76°, α0°, 106°, 130°, 156°, 180°, 1α6°, 210°, 236° and 260°.

What is interesting with this is that it clearly alternates between 6 and 0 on the last number. Where 76° and 1α6° stand out in notation on 8 it seems clearer on 12. What is also interesting is to see the mirroring pattern of 39° and 169° compared to ω3° and 223°. This noting to correlate to 3,6,9,10 (denary 12), due to its correlation of dividing 12 into 4.

In short, 8 division goes in a cadans of 9,6,3 and 0 while the 12 division goes in 6 and 0. On the tabel to the right we see the degrees and their fractions. I put the fractions only in denary, for the duodecimal vallue I would only have to change the 11 and the 12 to ω and 10.

Seemed redundant to do.



Duodecimal	Denary	
26°	30°	1/12
39°	45°	1/8
50°	60°	1/6
76°	90°	¼
α0°	120°	1/3
ω3°	135°	3/8
106°	150°	5/12
130°	180°	½
156°	210°	7/12
169°	225°	5/8
180°	240°	2/3
1α6°	270°	3/4
210°	300°	5/6
223°	315°	7/8
236°	330°	11/12
260°	360°	1/1

GDS Fractions.

Lets have a look at Denary fractions, first we have the main branches.

1. Traditional Fractional Classifications.
   - **Proper Fractions:** Fractions where the numerator (top) is smaller than the denominator (bottom), resulting in a value less than 1.
     - Examples: $\frac{1}{2}, \frac{3}{4}, \frac{5}{12}$
   - **Improper Fractions:** Fractions where the numerator is greater than or equal to the denominator, resulting in a value of 1 or greater.
     - Examples: $\frac{4}{3}, \frac{12}{5}, \frac{8}{8}$
   - **Mixed Numbers:** A combination of a whole number and a proper fraction.
     - Examples: $1\frac{1}{2}$, $3\frac{1}{4}$ (which is equal to $\frac{13}{4}$)
2. Positional Representations.
   - **Terminating Fractions:** Fractions that resolve cleanly and stop after a finite number of digits. In denary, this only occurs if the denominator's prime factors are 2 and 5.
     - Examples: $\frac{1}{2}$ =0.5, $\frac{1}{4}$=0.25
   - **Repeating / Recurring Fractions:** Fractions where the digits past the radix point repeat infinitely in a cyclical pattern because the denominator does not share clean factors with the base system.
     - Examples: $\frac{1}{3}$ =0.3333…, $\frac{1}{7}$=0.142857142857…
3. Special Classes and Relationships.
   - **Unit Fractions:** Fractions where the numerator is always exactly 1.
     - Examples: $\frac{1}{2}, \frac{1}{3}, \frac{1}{12}$
   - **Base-Power Fractions:** Fractions where the denominator is a clean power of the system's base. In denary, these are called *decimal fractions* (denominators of 10, 100, 1000).
     - Examples: $\frac{7}{10}$ =0.7, $\frac{25}{100}$=0.25
   - **Complex Fractions:** A fraction where the numerator, denominator, or both contain a fraction themselves.
     - Example: $\frac{1}{2} / \frac{3}{4}$
4. Relations between fractions.
   - **Fractions with a Common Denominator:** Separate fractions that share the exact same bottom value, making them instantly ready for basic arithmetic.
     - Example: $\frac{1}{8}$ and $\frac{3}{8}$
   - **Equivalent Fractions:** Fractions that utilize different numbers but represent the exact same physical value.
     - Example: $\frac{1}{2} = \frac{2}{4} = \frac{6}{12}$

Off course for most people this is known knowledge but still it seemed good to write down since we will look next at these same things in GDS to see the difference between duodecimal and denary.

1. Traditional Fractional Classifications. The structural definition of proper fractions, improper fractions, and mixed numbers remains completely unchanged, as these are universal mathematical concepts. However, the *notation* shifts to utilize the GDS symbols
   - **Proper Fractions:** Fractions where the numerator (top) is smaller than the denominator (bottom), resulting in a value less than 1.
     - Examples: $\frac{1}{2}$,and $\frac{3}{4}$ stay identical, but a fraction like ten-twelfths is cleanly written as $\frac{\alpha}{10}$.
   - **Improper Fractions:** Fractions where the numerator is greater than or equal to the denominator, resulting in a value of 1 or greater.
     - Example: A value like fourteen-twelfths transitions from denary $\frac{14}{12}$ to the pure GDS expression $\frac{12}{10}$.
   - **Mixed Numbers:** A combination of a whole number and a proper fraction.
     - Example: Mixed notation functions seamlessly; for instance, denary $1\frac{5}{12}$ is expressed in GDS as $1\frac{5}{10}$.
2. Positional Representations. This is where the most radical transformation occurs. Because twelve has more factors (2, 3, 4, 6) than ten (2, 5), the boundary between terminating and repeating fractions completely redraws itself.
   - **Terminating Fractions:** In GDS, any fraction whose denominator's prime factors are 2 and 3 will terminate cleanly. Consequently, denominators like 3, 4, 6, 8, and 9 all resolve into finite, clean numbers past the radix point.
     - Examples: $\frac{1}{3}$ =0.4, $\frac{1}{4}$=0.3, $\frac{1}{6}$=0.2, $\frac{1}{8}$=0.16 and $\frac{1}{9}$=0.14
   - **Repeating / Recurring Fractions:** The notorious infinite decimals of base-10 disappear for thirds and quarters. However, because 5 and 7 are not factors of twelve, fractions with these denominators will still repeat.
     - Examples: $\frac{1}{5}$ =0.247247…, $\frac{1}{7}$=0.186α35186α35…
3. Special Classes and Relationships.
   - **Unit Fractions:** Unit fractions remain highly vital, but their duodecimal expansion is far more practical for everyday divisions (halves, thirds, quarters, and sixths all become single-digit unit positions).
   - **Base-Power Fractions:** These completely shift from *decimal fractions* to *duodecimal fractions*. Instead of denominators being powers of ten (10, 100, 1000), they are now powers of twelve (10, 100, 1000) in GDS, representing 12, 144, and 1728).
     - Examples: $\frac{7}{10}$ =0.7 (which physically means seven-twelfths, or 58.33% in denary).
4. Relations between fractions.
   - **Fractions with a Common Denominator:** Because twelve can be divided into halves, thirds, and quarters without leaving any remainders, finding a common denominator for common daily measurements becomes significantly easier and requires much lower numbers than in base-10.
   - **Equivalent Fractions:** The mathematical equivalence chains naturally adapt to the new base grid, highlighting the clean alignment with the number twelve.
     - Example: $\frac{1}{2} = \frac{6}{10} = \frac{60}{100}$

GDS GMT.

One of the more interesting things of adjusting minutes to hours is with military time. Next I have made a table on how the hours are on a day. At first sight there is already one big improvement, that is that midday is defined as 10:00 instead of 12:00. So a 0 means before midday and a 1 after midday.

Now this is already something but when you look at minutes it becomes even weirder. Normally we stop at 59 and go to 00. This is of course since one hour is 60 minutes. But now with duodecimal we can count up to 120 units before needing to loop. That means that we can note 2 hours in minutes. I have noted that besides the normal GMT. It is still rather interesting that ternary GMT is now possible since that is a first in history and could be rather useful. It is noteworthy that the last minute before midnight would be noted as ω:ωω before going back to 0:00. I have also noted above the table if it is ternary or quaternary.

With more minutes in an hour I have noted the coinciding times. Time will in that regard feel bigger and faster with how fast the numbers will change over time. Still it is only a notation so the seconds will in practice move as fast as normal since the laws of physics do not break with this. It feels counterintuitive but perhaps it is useful in data compression rather than in daily life. Imagine how many terabytes such a time notation will save.

The only way how you could make time binary would be with Babylonian sexagesimal cuneiform.

Quaternary		Ternary
Denary GMT	GDS GMT	GDS GMT
01:00	01:00	0:60
02:00	02:00	1:00
03:00	03:00	1:60
04:00	04:00	2:00
05:00	05:00	2:60
06:00	06:00	3:00
07:00	07:00	3:60
08:00	08:00	4:00
09:00	09:00	4:60
10:00	0α:00	5:00
11:00	0ω:00	5:60
12:00	10:00	6:00
13:00	11:00	6:60
14:00	12:00	7:00
15:00	13:00	7:60
16:00	14:00	8:00
17:00	15:00	8:60
18:00	16:00	9:00
19:00	17:00	9:60
20:00	18:00	α:00
21:00	19:00	α:60
22:00	1α:00	ω:00
23:00	1ω:00	ω:60
00:00	00:00	0:00

Duodecimal expectations and implementations.

When I first needed to use duodecimal counting, I looked up what it is and how it works. Because most of it seemed highly impractical, I was prompted to make this for personal use. This resulted in a breach with how normally this topic is handled.
The first thing was that if I divide something over 12 but I have only 10 names how do I count with the other 2? Instead of having to type out some symbol that 99% of the world doesn't know about is highly inefficient and would make things look weird and alien.

So out of respect for western heritage where we use mostly Greek mathematics I landed on alpha and omega. It also sounded linguistically okay enough to use without much confusion. Even though the Greeks optimized eastern mathematics it is still the east where it first came from, they were the first to map out the stars that give us the first notion of time and dimensions.
And how about the 10?

Well that one must shift because it is simply the first number as whole unit and the other notes the amount from 0 to 9. But to call it dozen and changing my whole vocabulary was a bridge too far. I can implement Alph and Ohm but I am not going to call a ten a dozen if a ten is actually just the name for a whole unit. Sure, confusing at times but that is only because I use denary daily from a young age till now.

I was most surprised to see the unnecessary changes and requirements duodecimal notations required. People don't have to change their whole vocabulary to add two numbers to a system. The weirdest thing I came across was the idea to redefine the number of degrees in a circle from 360 to 144. Why? Because 12x12=144. This is absolutely absurd since not only 30x12=360. But also, since that way one degree would expand a lot. That would result in more decimals usage in degrees. To make an easy analogy it is compared to switching from 1080p to 240p on a video. The reverse is the reason why I have changed the minutes from denary to duodecimal as the hours are as well. The same thing is with the years but I already mentioned that part earlier in this document at chapter 5 (p6).

Duodecimal is tricky enough so I am against making it even more difficult. I even did the reverse in this document and tried to explain it as easy and clear as I can by providing visual presentations and hopefully analogies that are easy enough to understand. Duodecimal is a more efficient system as much as denary is also more efficient than binary. But there is not a perfect system.

For true mathematical perfection I would recommend learning Babylonian sexagesimal cuneiform. For other people learning 60 different symbols that correlate with amounts are impractical. The number of numeral bases one can understand correlates quite heavily with that person's intelligence. It is not for nothing that Babylonian sexagesimal cuneiform was mostly used by scribes, astronomical priests and royal family/elite people. Not just out of paranoia for uprisings but also because most people simply did not have the mental capacity to understand.

Luckily, we are not in the bronze age but in the intelligence age. Humanity has improved much and overall intelligence has also improved in most western and Asian countries. The upgrade to duodecimal is now way more doable than in ancient times. Not only our technology has improved much but also our way of understanding the universe.

Considering licensing. GDS is all public domain since you cannot patent number systems.

MadHatter out.